Using 10-K text to gauge COVID-related corporate disclosure

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Presented by **Ashok Kumar**

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Introduction

- COVID pandemic impacted global economy
- Corporate communication becomes critical during crisis
  - Shareholders are worried about their investments as they look for more info
  - Balance between transparency and containment of negative sentiments
  - Potential impact on stock return or risk of litigation

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Research Question

Examine how firms disclose firm-level information during COVID?

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Proposed Solution

Idea

Developed two measures to gauge intensity of COVID-related disclosure using firm's 10-K reports

- Overall COVID dictionary
- Contextual COVID dictionary (count of COVID related words/phrases preceded or followed by different tones)
  - To gauge how COIVD related disclosure is being expressed (eg, positively, negatively etc)

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Methodology: RQ-1

Data Collection and Parsing: MD&A focused

Preprocessing

Used Word2Vec to create dictionary

Generated scores for:

a) Overall COVID freq

b) Contextual COVID

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Data

- Collected 10-K reports for S&P listed companies from EDGAR
- Fiscal year 2020
- Parsed MD&A section or item

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Data Pre-Processing

- Eliminate emails, URLs, new line characters, extra spaces etc.
- NER removal
- Lemmatization
- Stop words removal
- Tokenization (for Phraser model and Word2Vec)

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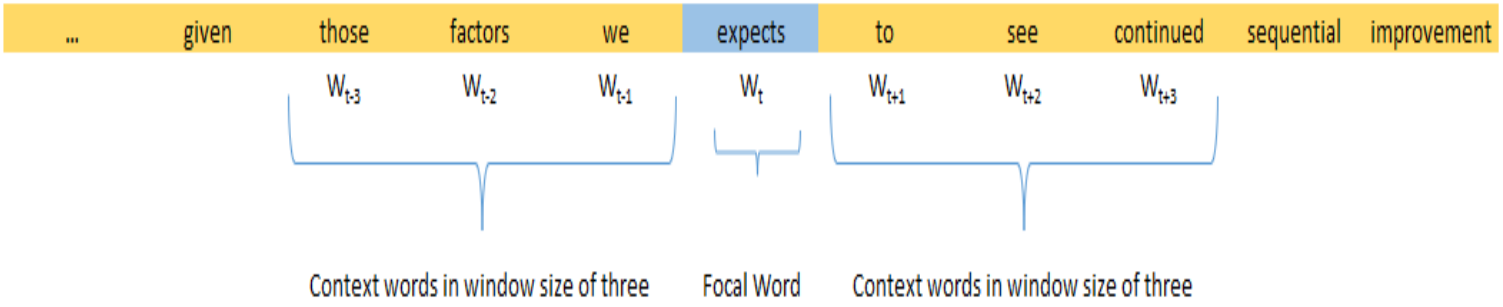
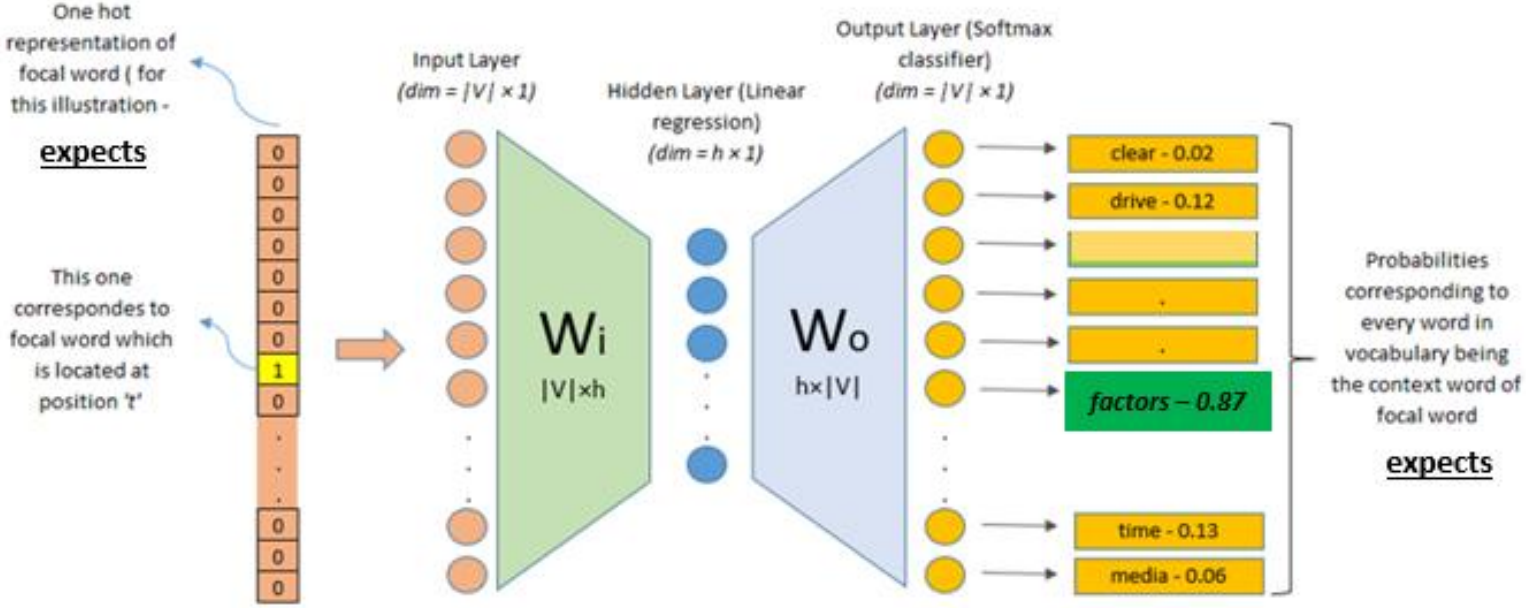
Word2Vec model





Developing COVID dictionary

Link to paper

Link to dictionary building doc file

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Question arises (contextual dict)

After developing COVID dictionary, how can we it use to measure COVID-related disclosure and contextual COVID disclosures?

- First, we simply count the frequency of COVID words
- For contextual capture, we employed WINDOW algorithm

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WINDOW algorithm

Link to paper

Link to doc file containing WINDOW algo examples

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COVID-related disclosure measures

4 measures:

1. frequency of **overall** COVID-related disclosure
2. contextual COVID-related **positive** word frequency
3. contextual COVID-related **negative** word frequency
4. contextual COVID-related **financial constraints** word frequency

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Regression setup: 1

Relationship between

- COVID-related disclosure (4 measures)

AND

- corporate liquidity events (measured by dividend payments and dividend change)

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Regression setup: 2

Relationship between

- COVID-related disclosure (4 measures)

AND

- corporate governance characteristics (measured by board independence and female directors)

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Results: setup 1

Table 5. Relationship between COVID-related disclosure and dividend payment.

	DV: Dividend payouts			
Variables	**Model (1)**	**Model (2)**	**Model (3)**	**Model (4)**
Frequency of COVID-related disclosure	-0.001***			
	(-2.985)			
Contextual COVID-related positive word frequency		-0.001		
		(-1.561)		
Contextual COVID-related negative word frequency			-0.001***	
			(-3.007)	
Contextual COVID-related financial constraints word frequency				-0.002***
				(-3.545)

- Model 1: High frequency of COVID-related words leads to lower dividend payouts (more uncertain firms -> likely to decrease their dividends)
- Model 2: Positive spin on the COVID-related disclosure means companies are better able to deal with COVID-related uncertainties and therefore, unlikely to affect dividend payments
- Model 3 & 4: Highlights that as the frequency of negative and financially constraining words in 10-K reports rises, the likelihood of a reduction in dividend payment increases

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Results: setup 1

Table 6. Relationship between COVID-related disclosure and net change in dividend payment.

	DV: Change in dividend payouts			
Variables	**Model (1)**	**Model (2)**	**Model (3)**	**Model (4)**
Frequency of COVID-related disclosure	-0.005***			
	(-3.461)			
Contextual COVID-related positive word frequency		-0.007***		
		(-3.605)		
Contextual COVID-related negative word frequency			-0.004***	
			(-4.523)	
Contextual COVID-related financial constraints word frequency				-0.009***
				(-4.285)

- Model 1: Suggests that a high frequency of COVID-related words reduces the possibility that dividend payments will increase
- Model 2, 3, & 4: It emphasizes that as the frequency of positive, negative, and financially constraining words in 10-K filings increases, the likelihood of a dividend increase lowers.

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Results: setup 2

Table 7. Relationship between governance characteristics and frequency of COVID-related disclosure.

	DV: Frequency of COVID-related disclosure	
Variables	**Model (1)**	**Model (2)**
Board independence%	0.922***	
	(2.769)	
Female director%		1.463***
		(3.140)

Model 1 & 2 present the relationship between the percentage of board independence, percentage of female directors, and overall COVID dictionary

- Results suggests that firms with more independent and female directors are likely to disclose more COVID-related information

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Results: setup 2

Table 8. Relationship between board-level characteristics and contextual COVID-related positive disclosure.

	DV: Contextual COVID-related positive word frequency	
Variables	**Model (1)**	**Model (2)**
Board independence%	0.695***	
	(3.717)	
Female director%		0.785***
		(3.269)

Model 1 & 2 shows the relationship between the percentage of board independence, percentage of female directors, and COVID related positive word frequency

- It highlights that corporate disclosure is likely to have amore positive tone amid the COVID-19 pandemic if there is a high level of board independence and female directors

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Results: setup 2

Table 9. Relationship between board-level characteristics and contextual COVID-related negative disclosure.

	DV: Contextual COVID-related negative word frequency	
Variables	**Model (1)**	**Model (2)**
Board independence%	1.300***	
	(2.641)	
Female director%		2.057***
		(3.284)

Model 1 & 2 shows the relationship between the percentage of board independence, percentage of female directors, and COVID-related negative word frequency

- These findings state that in the presence of more independent and female directors, corporate disclosure is likely to contain negative tones amid the economic and supply chain disruption caused by the COVID-19 pandemic → to reduce information asymmetry during a crisis

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Results: setup 2

Table 10. Relationship between board-level characteristics and contextual COVID-related financial constraint disclosure.

	DV: Contextual COVID-related financial constraints word frequency	
Variables	**Model (1)**	**Model (2)**
Board independence%	0.171	
	(0.855)	
Female director%		0.417
		(1.640)

Model 1 & 2 shows the relationship between the percentage of board independence, percentage of female directors, and COVID-related financial constraint word frequency

- Results are insignificant, implying that corporate governance does not have much influence in disclosing financially constraining news at the time of COVID-19

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Result summary

- Overall COVID dictionary effectively predicts a firm's liquidity event
- Similar results for contextual COVID dictionaries with a negative spin
- Results suggest that better-governed firms prefer greater transparency
- Our results further show that better-governed firms (e.g., greater board independence, and more female directors) tend to have more COVID-related disclosures, even though more COVID related disclosures suppress a firm's market-based stock performance (e.g. Tobin's Q)

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Navigating in Turbulent Times: Using Social Media to Examine Small and family-Owned Business Topics and Sentiments during the COVID-19 Crisis

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Presented by **Ashok Kumar**

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Introduction

- Small and family-owned businesses tend to experience more severe economic consequences than their larger counterparts
- Often lack financial resources needed to weather the challenges
- No formal communication channels

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Research Question: 1

What topics pertaining to small and family-owned businesses do social media users discuss during the COVID-19 pandemic?

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Research Question: 2

What is the impact of major government announcements on these discussions?

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Proposed solution

RQ1:

- We employed the BERTopic model, a SOTA technique for topic modeling

RQ2:

- We study how sentiments change around a major government announcement aimed at supporting small businesses in the face of the pandemic

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BERTopic Implementation

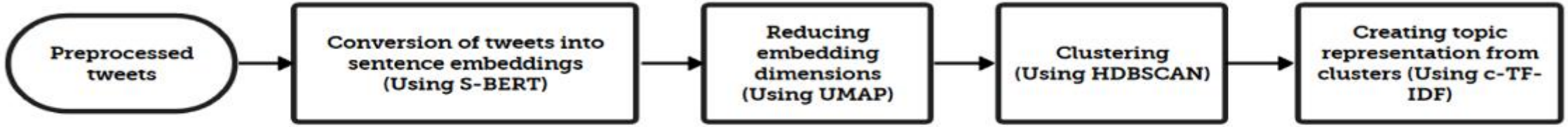

Fig. 2 BERTopic implementation stages

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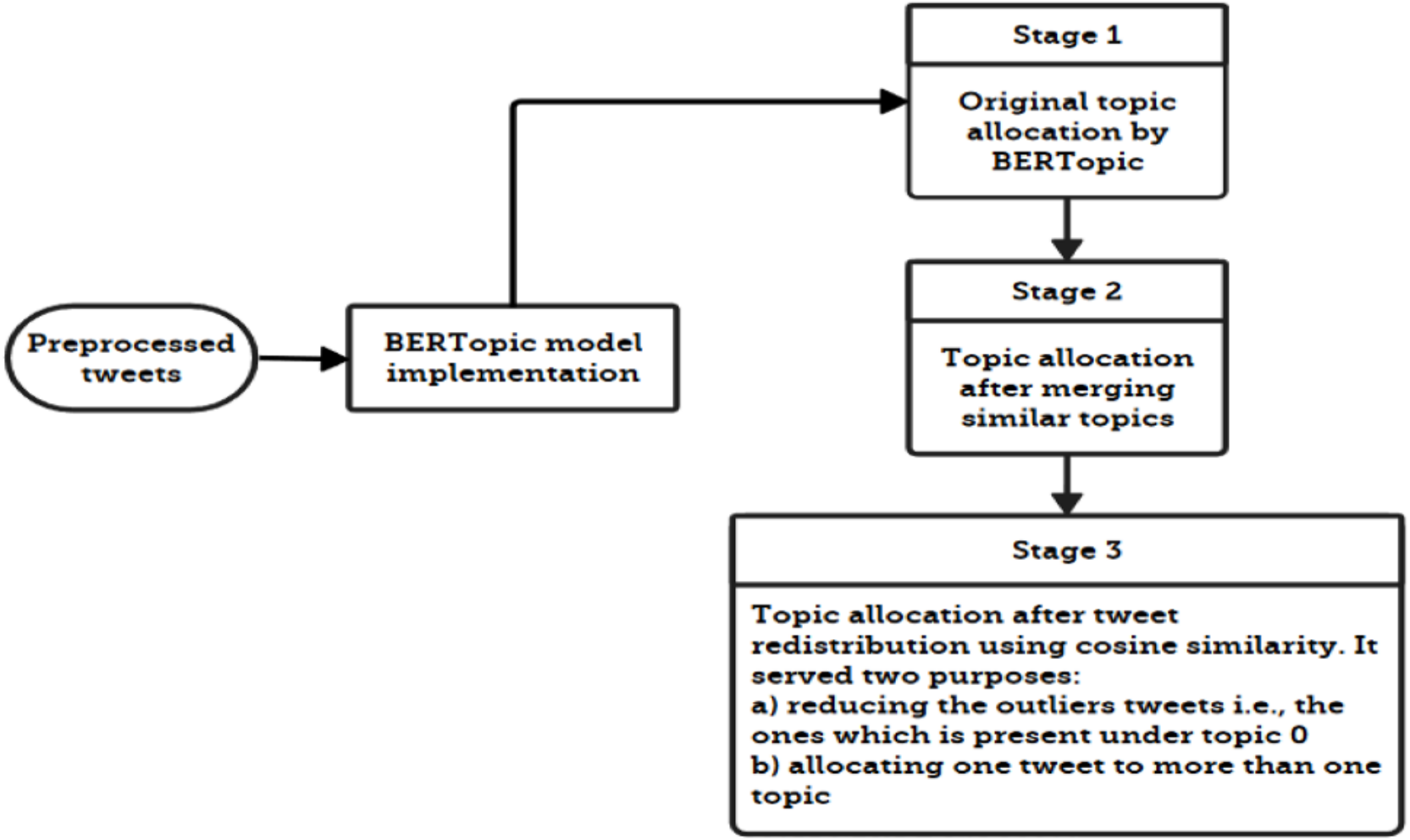
Topic allocation stages



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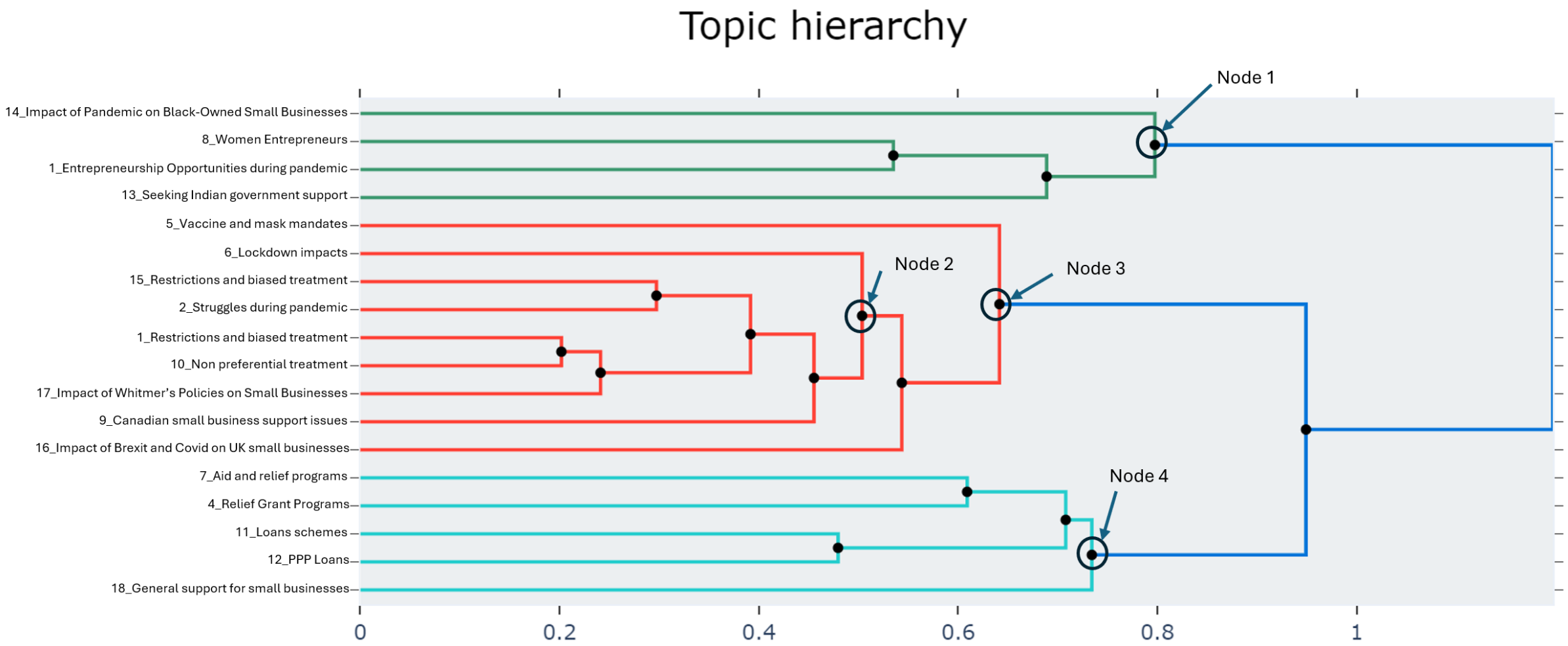
Topic allocation stages



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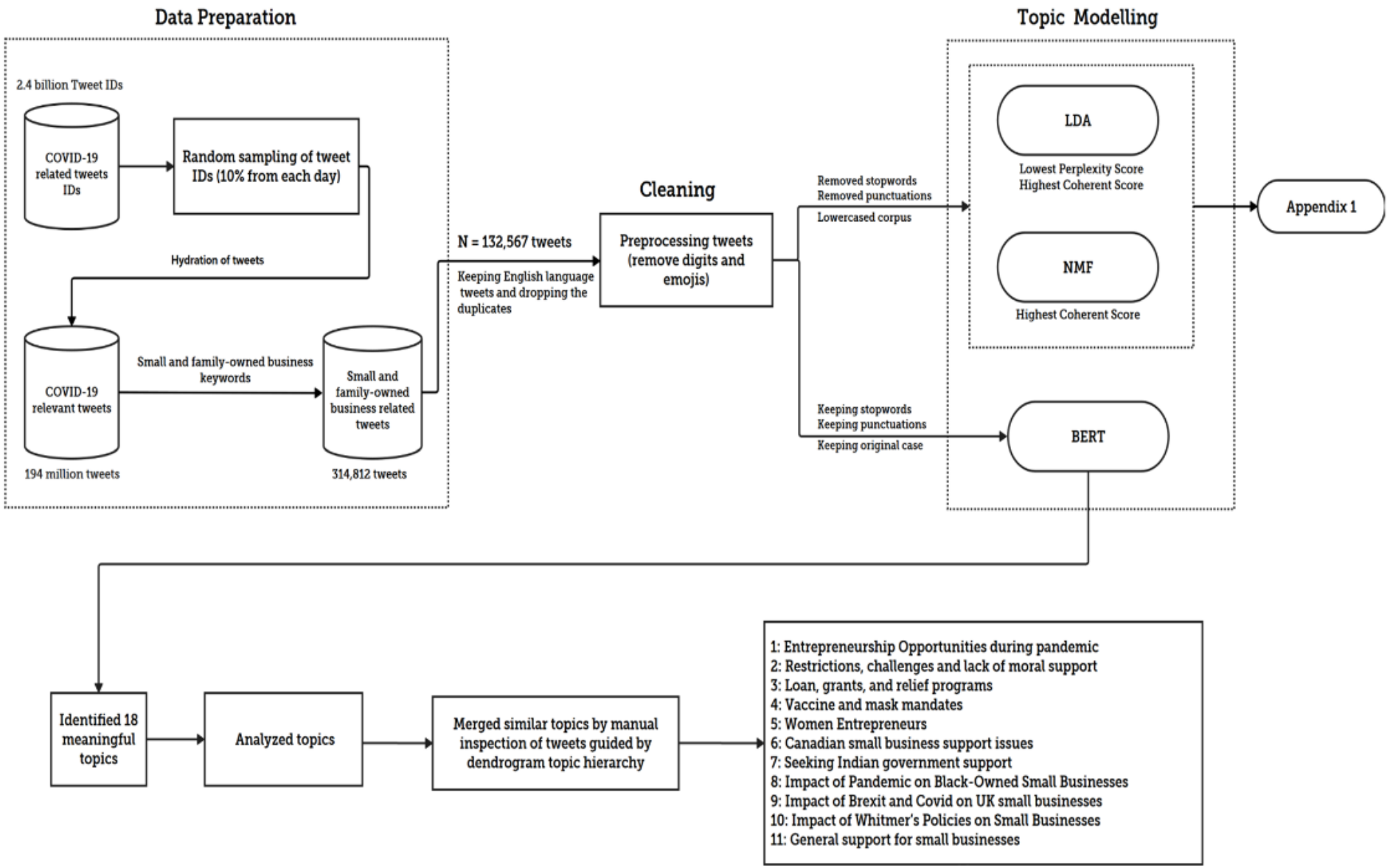
Complete Process model



Results

- Took final topic allocation topic distribution
- Identified key announcements and their dates
- Temporal window selection (15 days and 20 days)
- Generated LIWC features (tone_neg, emo_anx) around announcements

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Results: Topics

Topic number	Topic	Description	Top key-words figure (Stage 3 Word Clouds)
1	Entrepreneurship Opportunities during pandemic	Discussions on the role of new entrepreneurs in economic recovery and the opportunities that arise by the pandemic.	Figure 4
2	Restrictions, challenges and lack of moral support	Discussions on the impact of lockdowns and restrictions on small businesses, the preferential treatment of large corporations, and the struggles and lack of support faced by small businesses during the pandemic.	Figure 5
3	Loan, grants, and relief programs	Discussions on the financial aid, relief programs, and loan schemes available to small businesses during the COVID-19 pandemic.	Figure 6
4	Vaccine and mask mandates	Discussions of views on COVID-19 mandates such as vaccinations and masks	Figure 7
5	Women Entrepreneurs	Discussions on the hurdles faced by the women entrepreneurs.	Figure 8
6	Canadian small business support issues	Issues and concerns over the support provided to the Canadian small businesses	Figure 9

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Results: Topics

7	Seeking Indian government support	Tweets discussing the impacts of the lockdown on the Indian small businesses and the need for the support from the government.	Figure 10
8	Impact of Pandemic on Black-Owned Small Businesses	Tweets underscoring the lack of support and challenges faced by the black owned businesses.	Figure 11
9	Impact of Brexit and Covid on UK small businesses	Discussions around the effects of the pandemic and Brexit on UK small businesses.	Figure 12
10	Impact of Whitmer's Policies on Small Businesses	Discussions around the criticism directed towards the inadequacy of Whitmer's policies on small businesses during the pandemic.	Figure 13
11	General support for small businesses	This topic involves discussions of general support for small business rather than being more focused discussions like topic 12 where discussions revolve around PPP loans.	Figure 14

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Key announcements

Table 3 Key announcements

Announcement	Date	Description
Announcement 1: Fed launched the Paycheck Protection Program Liquidity Facility (PPPLF)	9th April 2020	To help small businesses, the Fed launched the Paycheck Protection Program Liquidity Facility (PPPLF) on April 9, 2020, in concert with the CARES Act. This program lent money to banks so they could, in turn, lend money to small businesses through the Paycheck Protection Program (PPP) (Board of Governors of the Federal Reserve System, 2020, April 9)
Announcement 2: Emergency Rent Subsidy program announced	9th October 2020	The Canada Emergency Rent Subsidy Program (CERS). This program consists of two components: the base rent subsidy, accessible to organizations facing ongoing revenue decline, and the Lockdown Support, which offers supplementary assistance to entities forced to shut down or considerably curtail operations due to public health orders (Department of Finance Canada, 2020, September 10)
Announcement 3: Chancellor Sunak announced a £4.6bn fresh financial support package for struggling	5th January 2021	Chancellor Sunak unveiled a new financial assistance initiative amounting to £4.6 billion, aimed at providing vital support to struggling companies in the UK. This comprehensive package comprises two key components which aims to provide significant financial relief to a wide range of businesses across various sectors, addressing the challenges posed by the ongoing economic disruptions (The Guardian, 2021, January 5)
Announcement 4: SBA increased the maximum loan amount from $150,000 to $500,000	24th March 2021	SBA increased the maximum loan amount from $150,000 to $500,000 (U.S. Small Business Administration, 2021, March 24)
Announcement 5: SBA increased the COVID EIDL maximum loan amount from $500,000 to $2 million	9th September 2021	SBA increased the COVID EIDL maximum loan amount from $500,000 to $2 million (U.S. Small Business Administration, 2021, September 9)

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Results: t-test

Table 4 Announcement 1 t-test results

		Panel A: 15 days window			Panel B: 20 days window		
topic	liwc_sig	*n* (post)	*n* (pre)	mean_diff (post - pre)	*n* (post)	*n* (pre)	mean_diff (post - pre)
1	tone_neg	103	80	-1.803***	137	89	-1.313**
1	emo_anx	103	80	-0.327*	137	89	-0.257*
2	tone_neg	151	119	0.306	210	158	0.403
2	emo_anx	151	119	-0.058	210	158	-0.079
3	tone_neg	304	202	0.936***	404	249	0.778***
3	emo_anx	304	202	0.074	404	249	0.098**
5	tone_neg	10	8	6.776	11	10	6.419
5	emo_anx	10	8	0	11	10	0
8	tone_neg	9	6	-1.687	19	6	-2.198
8	emo_anx	9	6	0	19	6	0
11	tone_neg	29	43	0.027	45	57	0.697
11	emo_anx	29	43	-0.241	45	57	-0.182

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Results: t-test

Table 5 Announcement 2 t-test result

		Panel A: 15 days window			Panel B: 20 days window		
topic	liwc_sig	*n* (post)	*n* (pre)	mean_diff (post - pre)	*n* (post)	*n* (pre)	mean_diff (post - pre)
6	tone_neg	56	31	-0.941	73	40	-1.121*
6	emo_anx	56	31	0.056	73	40	-0.025

Table 6 Announcement 3 t-test results

		Panel A: 15 days window			Panel B: 20 days window		
topic	liwc_sig	*n* (post)	*n* (pre)	mean_diff (post - pre)	*n* (post)	*n* (pre)	mean_diff (post - pre)
9	tone_neg	40	22	0.043	48	28	-0.324
9	emo_anx	40	22	0.128	48	28	0.013

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Results: t-test

Table 7 Announcement 4 t-test results

		Panel A: 15 days window			Panel B: 20 days window		
topic	liwc_sig	*n* (post)	*n* (pre)	mean_diff (post - pre)	*n* (post)	*n* (pre)	mean_diff (post - pre)
1	tone_neg	192	248	0.136	276	311	0.376
1	emo_anx	192	248	-0.054	276	311	-0.010
2	tone_neg	471	434	-0.068	627	564	-0.026
2	emo_anx	471	434	-0.060	627	564	-0.068
3	tone_neg	211	338	0.451*	252	525	0.252
3	emo_anx	211	338	-0.148***	252	525	-0.105**
5	tone_neg	70	130	0.703	86	223	0.536
5	emo_anx	70	130	0.021	86	223	0.021
8	tone_neg	35	38	1.183	39	50	0.105
8	emo_anx	35	38	0	39	50	0
11	tone_neg	10	15	-0.919	13	20	-1.909
11	emo_anx	10	15	-0.215	13	20	-0.161

Table 8 Announcement 5 t-test results

		Panel A: 15 days window			Panel B: 20 days window		
topic	liwc_sig	*n* (post)	*n* (pre)	mean_diff (post - pre)	*n* (post)	*n* (pre)	mean_diff (post - pre)
1	tone_neg	142	129	-0.442	176	190	-0.371
1	emo_anx	142	129	0.016	176	190	0.071
2	tone_neg	286	252	0.230	362	332	0.293
2	emo_anx	286	252	-0.070	362	332	-0.056
3	tone_neg	94	83	0.318	119	112	0.395
3	emo_anx	94	83	-0.006	119	112	0.032
5	tone_neg	46	28	-0.253	51	37	-0.526
5	emo_anx	46	28	0.063	51	37	0.057
8	tone_neg	16	16	1.945	23	19	1.204
8	emo_anx	16	16	0	23	19	0.248
11	tone_neg	4	2	3.120	6	2	1.630
11	emo_anx	4	2	0.675	6	2	0.450

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Results summary

- Findings suggest that government announcements did not change the negative sentiments for most of the topics – highlighting the ineffectiveness of government announcements in alleviating people's concern related to small and family-owned business

- For e.g.,
  - For topic 3, A1 saw an increase in both negative emotions at 1% level, meaning they gets even stronger post announcement!
  - A3 and A5 had no significant impact on negative tone or anxiety levels across all topics

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Thank You!